

# **Experiment Title: Static and Dynamic Routing in Computer Networks**

## **Objective**

- To understand the concept of routing in computer networks.
- To configure and compare static routing and dynamic routing (RIP).
- To analyze the behavior, advantages, and limitations of both routing methods.
- To verify network connectivity between multiple LANs using routers.

## **Introduction**

Routing is a fundamental process in computer networks that determines how data packets are forwarded from a source to a destination across different networks. Routers use routing tables to decide the best possible path for packet delivery. There are two main types of routing techniques: static routing and dynamic routing. Static routing requires manual configuration of routes by the network administrator, while dynamic routing uses routing protocols to automatically learn and update routes. This experiment demonstrates the configuration and working principles of both routing methods using Cisco routers.

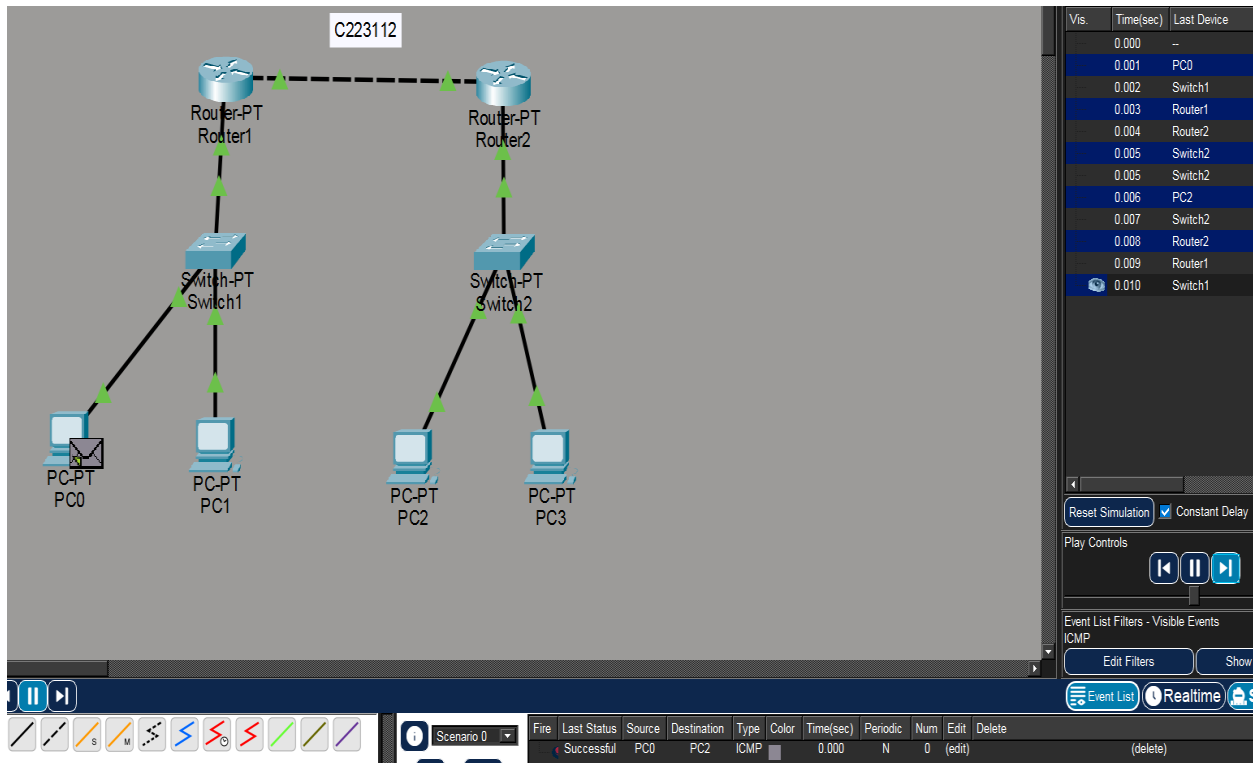
## **Equipment & Tools**

- Cisco Packet Tracer software
- Two Cisco routers
- Two switches
- Four PCs
- Ethernet and serial cables
- Command Line Interface (CLI)

## **Experimental Setup**

The network topology consists of two routers connected to each other. Each router is connected to a switch, and each switch connects two PCs. IP addresses are assigned to all devices according to their respective networks. In the static routing part, routes are manually configured using the ip route command. In the dynamic routing part, OSPF & RIP version 2 is enabled on both routers to allow automatic exchange of routing information.

**STATIC ROUTING:** Static Routing: Static routing is manually configured by the administrator. Routes must be added individually for each destination network. It is simple but not suitable for large networks.



```
Router1
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fastethernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int fa1/0
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
exit
Router(config)#ip route 192.168.20.0 255.255.255.0 192.168.30.2
Router(config)#exit
Router#
```

```
Router2
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fastethernet0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

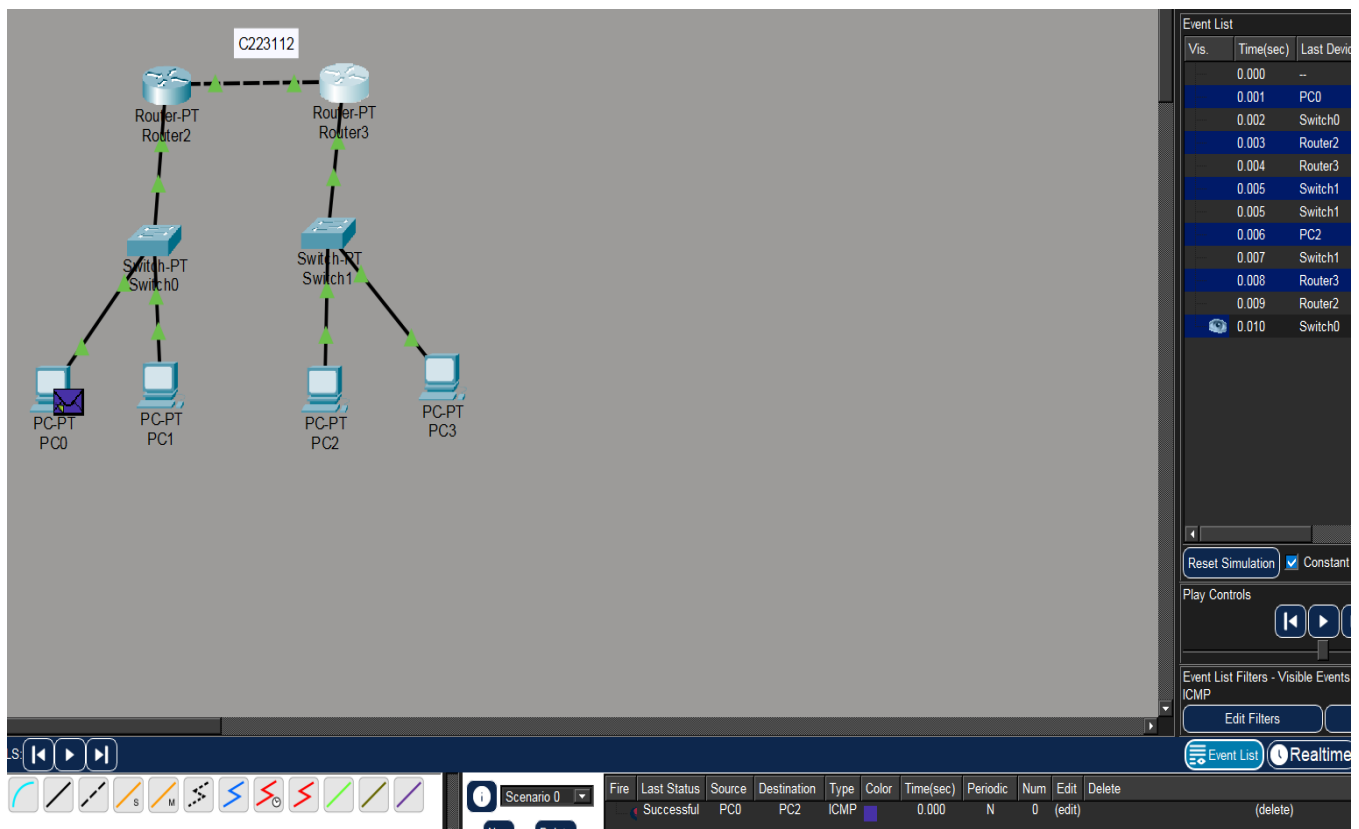
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int fa1/0
Router(config-if)#ip address 192.168.30.2 255.255.255.0
Router(config-if)#no shut
```

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## DYNAMIC ROUTING:

## OPCF:



The image shows two terminal windows from Cisco Packet Tracer. The left window is for Router2, and the right window is for Router3. Both show the configuration of interfaces, IP addresses, and OSPF settings.

```
Router2
Router(config)#
Router(config)#
Router(config)#interface fastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int fa0/1
%Invalid interface type and number
Router(config)#interface fa1/0
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown

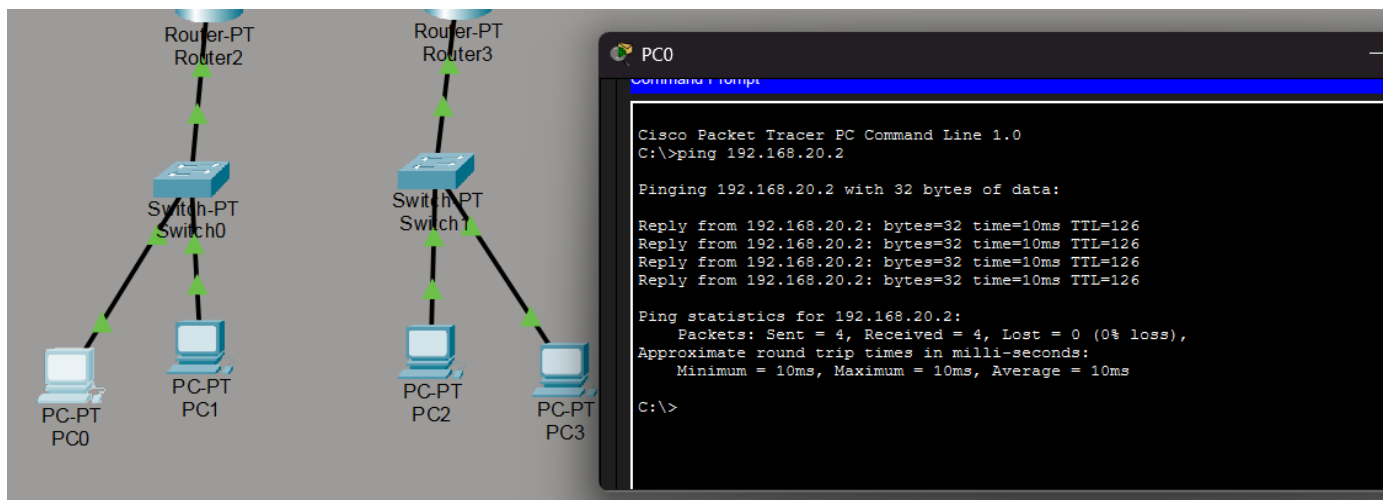
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.10.0 0.0.0.255 area 1
Router(config-router)#network 192.168.30.0 0.0.0.255 area 1
Router(config-router)#exit
Router(config)#

Router3
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastEthernet0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#int fa1/0
Router(config-if)#ip address 192.168.30.2 255.255.255.0
Router(config-if)#no shutdown

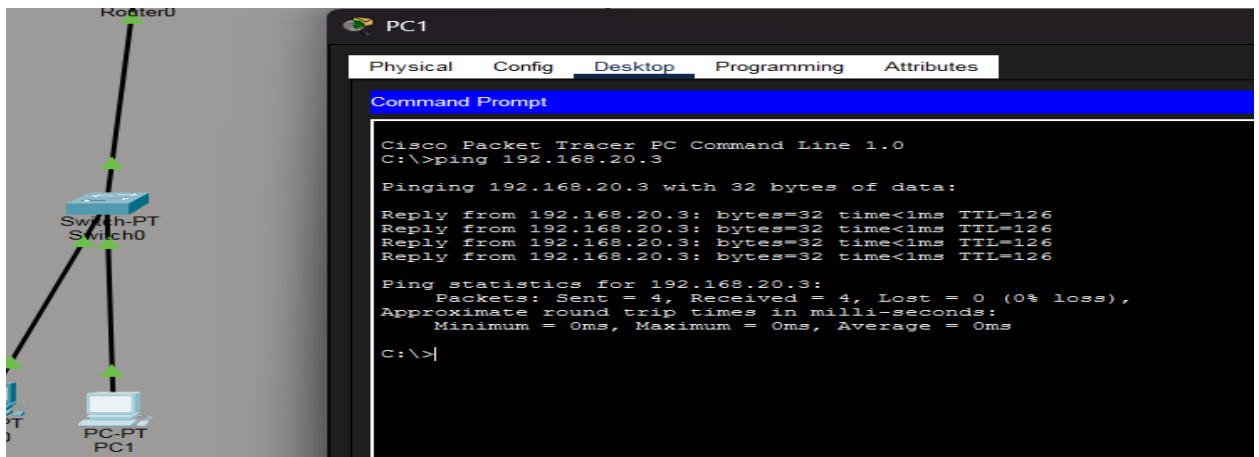
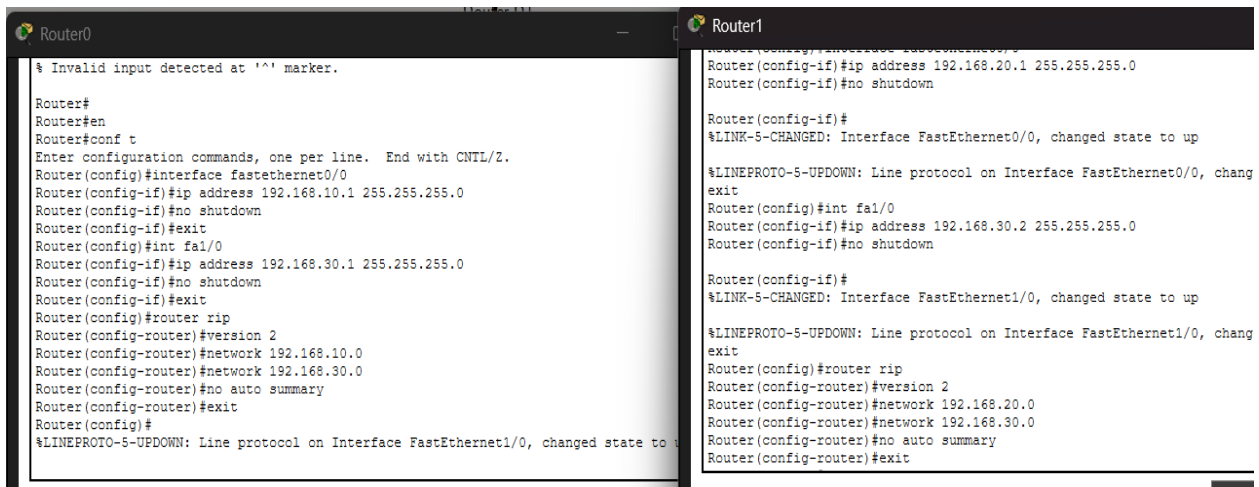
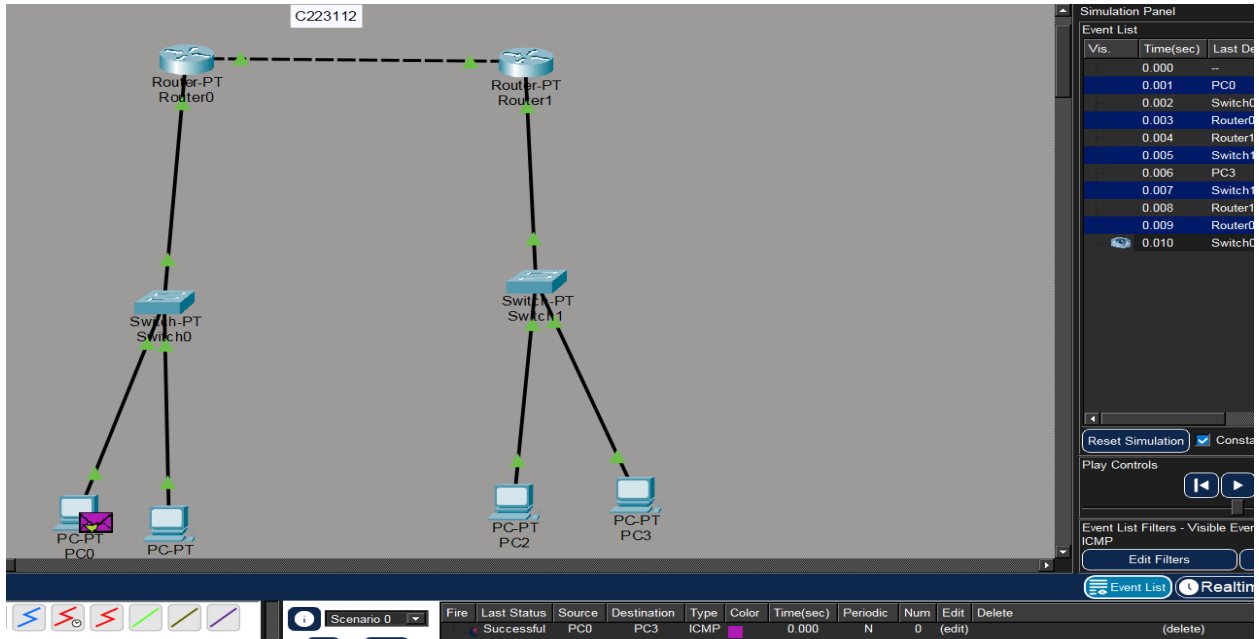
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.20.0 0.0.0.255 area 1
Router(config-router)#network 192.168.30.0 0.0.0.255 area 1
Router(config-router)#exit
```



All packets reached their destination.

After enabling OSPF, routers automatically learned all networks. shows messages are getting sent successfully.

## RIP VERSION 2:



## Result and Analysis

After configuring static routing, successful communication was observed between PCs located in different networks, proving correct manual route configuration. In dynamic routing using RIP, routers automatically learned routes to remote networks, which was verified using the `show ip route` command. Dynamic routing reduced manual effort and adapted better to network changes compared to static routing.

## Discussion

This experiment helped to understand how static and dynamic routing work in a simple network. Static routing was easy to configure but required manual route entry on each router. Dynamic routing using RIP was more flexible, as routers automatically learned routes and shared information. Overall, dynamic routing was easier to manage than static routing.

## Problems Faced:

- Interfaces were down due to missing `no shutdown` command.
- Incorrect IP address or default gateway on PCs.
- Initial confusion with router interfaces and connections.